

1 ALETIA INDEX – AI/ML INTELLIGENCE PROBLEM SUMMARY
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3 Written: March 2026
4
5 THE CORE PROBLEM
6 -----
7 The FDA's classification system was built for hardware risk, not software intelligence.
8 AI/ML is being shoehorned into it. The regulatory system doesn't have a clean, mandatory
9 "this is AI/ML" flag. Manufacturers self-describe, and the FDA retroactively tags devices.
10
11 A 510(k) for an AI device says:
12 "This algorithm, trained on this data, performed this well on this test set, at this point in time."
13
14 It says nothing about:
15 - What happens when the algorithm drifts because the hospital population changes
16 - Whether the device has been validated on diverse demographics
17 - Whether there's any post-market monitoring at all
18 - Whether the clearance reflects the current version of the software
19
20 WHAT WE LEARNED FROM THE DATA
21 -----
22 1. Product code expansion from 6 to 57 codes caught many more devices, but still catches
23 devices by CATEGORY rather than by AI NATURE. Some of the 5,037 rows are hardware
24 devices that happen to share a product code with AI software.
25
26 2. The inverse is also true – devices like K251293 (CardioVision / QUO) are clearly AI/ML
27 but sit in a product code that doesn't signal it. These will keep appearing.
28
29 3. After seeding: ~5,037 devices. A significant number are MRI machines and ultrasound
30 systems with an incidental AI image enhancement feature – not what clinicians or
31 procurement officers are looking for on Aletia.
32
33 4. Surfacing hardware-dominant devices equally with dedicated AI diagnostic tools
34 undermines the index's clinical credibility.
35
36 THE MEANINGFUL DISTINCTION
37 -----
38 The important difference between Software as a Medical Device (SaMD) and AI/ML devices
39 is the ONGOING COMPLIANCE BURDEN beyond just initial clearance.
40
41 AI/ML devices are challenged by their newness:
42 - Locked algorithms cleared in 2019 may no longer reflect current performance
43 - Continuously learning systems require different oversight entirely
44 - Post-market surveillance frameworks weren't designed for adaptive software
45 - Cleared doesn't mean currently performing (ACR's Assess-AI addresses exactly this)
46
47 DIMENSIONS THAT MATTER CLINICALLY (not captured publicly anywhere)
48 -----

49 1. MODEL TRANSPARENCY
50 Does the manufacturer disclose training data demographics, model architecture,
51 known failure modes? FDA doesn't require this to be public.
52 Aletia could make it a field and reward disclosure with a higher trust signal.
53

54 2. CONTINUOUS LEARNING FLAG
55 Is this a locked algorithm or does it update?
56 A locked algorithm cleared in 2019 is a fundamentally different risk profile
57 to a continuously learning system cleared under a Predetermined Change Control Plan (PCCP).
58 The ai_ml_type field (Locked / Continuous Learning) exists but is not being populated.
59

60 3. VALIDATION POPULATION MATCH
61 Was this device validated on data that resembles the South African population?
62 Almost never disclosed. Major clinical risk for African deployment.
63 Nobody is surfacing this. Aletia could.
64

65 4. REAL-WORLD PERFORMANCE EVIDENCE
66 Is there any published peer-reviewed evidence beyond the clearance study?
67 The clinical_audits table touches this but is manually populated.
68

69 5. REGULATORY PATHWAY AS A SIGNAL
70 De Novo devices are genuinely novel.
71 510(k) devices claiming predicate on a 2015 device may be stretching
72 substantial equivalence significantly.
73 The pathway tells you something about how hard the FDA looked.
74

75 PROPOSED TIERED APPROACH
76 -----
77 Rather than being comprehensive (all 5,037 devices with basic fields),
78 Aletia's value is in being SELECTIVELY DEEP.
79

80 TIER 1 – Registry Listings (current state)
81 Auto-populated from FDA/MHRA/EUDAMED. Basic fields, health status, jurisdiction flags.
82 Exists to be found via SEO and give manufacturers a reason to claim.
83

84 TIER 2 – Claimed and Enriched Listings
85 Manufacturer has verified and added:
86 - Model transparency data
87 - Validation demographics
88 - PCCP status
89 - Real-world evidence links
90 This is the paid product. Meaningfully different from a registry entry.
91

92 TIER 3 – Aletia Verified
93 Human clinical audit completed. 10-point checklist. Badge awarded.
94 The premium trust signal. Cannot be self-assigned.
95

96 WHAT TO BUILD NEXT (in order)

97 -----

98 1. ai_ml_integral BOOLEAN

99 Is the AI/ML function the PRIMARY clinical purpose, or incidental?

100 Devices where it's incidental get a minimal listing.

101 Devices where it's integral get the full treatment.

102 This is the most important filter.

103

104 2. pccp_flag BOOLEAN

105 Does this device have an FDA-approved Predetermined Change Control Plan?

106 Publicly disclosed in the 510(k) summary for 2024+ devices.

107 Single best indicator the manufacturer is thinking seriously about

108 ongoing compliance burden.

109

110 3. validation_demographics TEXT

111 Free text initially, structured later.

112 Even noting "US-only validation data" vs "multi-site including African populations"

113 is clinically meaningful for the SAHPRA-focused audience.

114

115 4. SMARTER EXCLUSION WORKFLOW

116 Currently exclusion is entirely manual.

117 A simple admin rule engine that auto-flags devices where:

118 - Product code is hardware-dominant (LNH, IYN, JAK)

119 - AND device name contains no AI/ML terms

120 ...would let you review 50 candidates at a time rather than finding them one by one.

121

122 THE KEY INSIGHT

123 -----

124 Aletia's value isn't being a bigger version of the FDA list.

125 It's being the place where the AI/ML NATURE of a device is actually interrogated

126 rather than assumed.

127

128 A database that contains both cleared devices AND pre-approval startups becomes

129 the most comprehensive view of the AI medical device landscape available anywhere –

130 the kind of dataset that justifies API licensing to EHR vendors, insurers, and

131 health policy researchers as a longer-term revenue stream.

132

133 The pre-clearance listings are what makes the data interesting to those buyers,

134 not just the cleared device index alone.

135